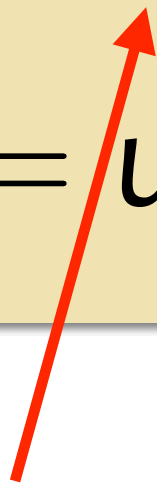


Order reduction and
weak stage order
for Runge-Kutta methods

Runge-Kutta methods

$$u'(t) = F(t, u)$$

$$u(0) = u^0$$



Often comes from PDE semi-discretization; includes boundary conditions

$$y^i = u^n + h \sum_{j=1}^m a_{ij} F(t_n + hc_j, y^j)$$
$$u^{n+1} = u^n + h \sum_{j=1}^m b_j F(t_n + hc_j, y^j)$$

c	A
	b^T

Order reduction

- Rate of convergence may be less than design order p
- Caused by:
 - Stiffness
 - Algebraic constraints (DAEs)
 - Time-dependent boundary conditions
- Rate of convergence is governed by stage order q
 - Typically we observe, at most:
 - Order q convergence for stiff ODEs
 - Order $\min(p, q+1)$ convergence for IBVPs

Stage error propagation

$$\tilde{y}^j = \tilde{u}^n + h \sum_{j=1}^m a_{ij} F(t_n + hc_j, \tilde{y}^j) + r_i$$

$$\tilde{u}^{n+1} = \tilde{u}^n + h \sum_{j=1}^m b_j F(t_n + hc_j, \tilde{y}^j) + r_{m+1}$$

Sources of perturbations:

- Rounding errors
- Implicit solve errors
- Spatial truncation error
- Imposition of boundary conditions

$$u'(t) = \lambda u$$

$$z = h\lambda$$

$$\varepsilon^{n+1} = P(z)\varepsilon^n + \sum_{k=2}^p h^k u^{(k)}(t_n) \underbrace{zb^T(I - zA)^{-1}}_{Q(z)} \underbrace{\frac{1}{k!} (c^k - kAc^{k-1})}_{\tau_k} + \mathcal{O}(h^{p+1})$$

Stage error propagation

$$\varepsilon^{n+1} = P(z)\varepsilon^n + \sum_{k=2}^p h^k u^{(k)}(t_n) b^T (I - zA)^{-1} \tau_k + \mathcal{O}(h^{p+1})$$

Conditions for order p
ensure that

$$b^T (I - zA)^{-1} \tau_k = \mathcal{O}(z^{p+1})$$

so we see convergence of
order p as $z \rightarrow 0$.

The conditions for order p
ensure also that

$$b^T \tau_k = 0$$

for $k \leq p$.

But for PDEs, we cannot
assume $z \rightarrow 0$, and we
need that

$$b^T (I - zA)^{-1} \tau_k = 0$$

for $k \leq p$.

See e.g. Sanz-Serna, Verwer,
& Hundsdorfer 1986

Stage order

Think of stages as approximate solutions:

$$y^i \approx u(t_n + hc_i)$$

Stage order residuals measure by how much each stage fails as an approximation of order k :

$$\tau_k = \frac{1}{k!} c^k - \frac{1}{(k-1)!} A c^{k-1}$$

A method has stage order q if:

$$\left. \begin{array}{l} \tau_k = 0 \\ \frac{1}{k} - b^T c^{k-1} = 0 \end{array} \right\} \text{ for all } k \leq q$$

Clearly, if the stage order is equal to the order, then order reduction is avoided.

But this is stronger than simply $b^T (I - zA)^{-1} \tau_k = 0$

Stage order of DIRKs and SDIRKs

- Maximal stage order of SDIRKs and explicit RKs: 1
 - Proof: first stage is backward (or forward) Euler
- DIRKs can have stage order 2, but only if first stage is explicit
 - Since then second stage is a combination of forward and backward Euler
- Example: Implicit trapezoidal rule

0	0	0
1	1/2	1/2
	1/2	1/2

$$\begin{aligned}y_1 &= u^n \\y_2 &= u^n + \frac{h}{2} (f(u^n) + f(u^{n+1})) \\u^{n+1} &= y_2\end{aligned}$$

An example

Diffusion equation with
time-dependent
boundaries:

$$\begin{aligned}u_t &= u_{xx} - \sin(t) \\ u(0, t) &= \cos(t) \\ u(1, t) &= \cos(t) \\ u(x, 0) &= 1\end{aligned}$$

Scheme	Order	Stage order	Convergence rate
Backward Euler	1	1	1
Implicit trapezoidal	2	2	2
DIRK ₃	3	1	2
DIRK ₄	4	1	2
Gauss6	6	3	4

Existing techniques for dealing with order reduction

- Use fully-implicit methods with high stage order (may be very costly for PDEs)
- Use linear multistep methods
- For boundary conditions:
 - Various modifications to the way they are implemented can alleviate order reduction
 - Sometimes at the cost of stability

Weak stage order

Recall that we need

$$b^T (I - zA)^{-1} \tau_k = 0$$

The conditions for order p ensure already that

$$b^T \tau_k = 0$$

for $k \leq p$.

Now what if τ_k is an eigenvector of A ?

A method of order $p > q$ has weak stage order q if the vectors τ_k are eigenvectors of A for all $k \leq q$.

More generally:

A method of order $p > q$ has weak stage order q if the vectors τ_k for all $k \leq q$ lie in an A -invariant subspace.

WSO introduced in Rosales, Seibold, Shirokoff, & Zhou

Roots of weak stage order

- The approach is based on ideas appearing in the works of Scholz (1989), Ostermann & Roche (1992, 1993), Alonso-Mallo & Palencia (2003), Hundsdorfer (1986), Skvortsov (2006, 2010, 2017), and Rang (2016)

Weak stage order in terms of Krylov spaces

$$Y = \text{span}\{b, A^T b, \dots, (A^T)^{s-1} b\}$$

$$K_q = \text{span}\{\tau_1, A\tau_1, \dots, A^{s-1}\tau_1, \tau_2, A\tau_2, \dots, A^{s-1}\tau_2, \dots, A^{s-1}\tau_q\}$$

The weak stage order is the largest value q such that these two spaces are orthogonal.

$$\dim(Y) + \dim(K_q) \leq s$$

Corollary 3.4. *For an s -stage Runge-Kutta scheme with order $p \geq 1$ and weak stage order q (with K_q and $P(x)$ defined in (15) and (20)), one has*

$$\deg(P) \leq \dim(K_q) \leq \begin{cases} s - \left\lfloor \frac{p+1+\sigma}{2} \right\rfloor, & \text{for fully implicit schemes,} \\ s - p - 1 + \sigma, & \text{for diagonally implicit schemes,} \end{cases}$$

where $\sigma = 1$ if A is invertible and the method is stiffly accurate, and $\sigma = 0$ otherwise.

DIRK methods with high weak stage order

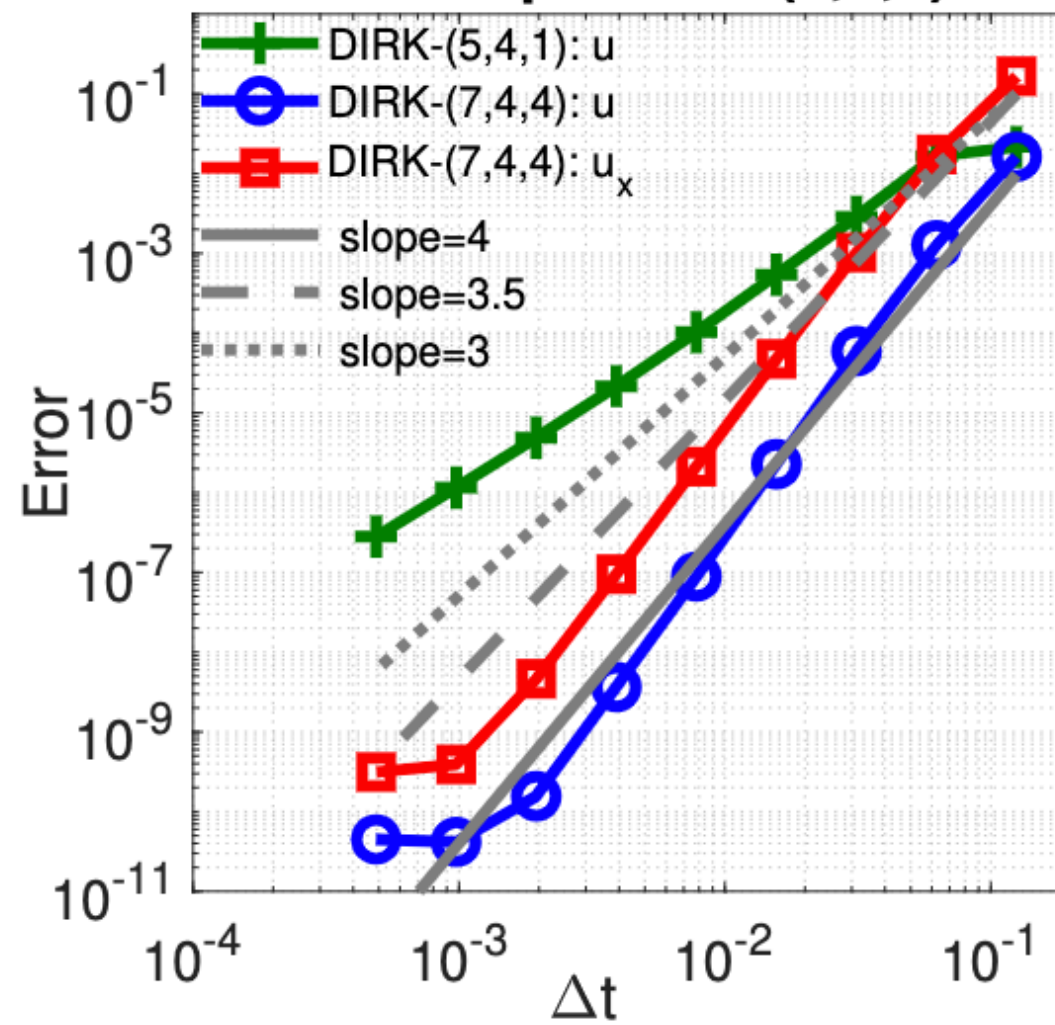
Order 3, WSO 3, L-stable:
(DIRK334)

0.138		0.138					
0.802		0.567	0.235				
2.332		-1.084	2.966	0.449			
1.000		0.598	-0.434	-0.053	0.890		
			0.598	-0.434	-0.053	0.890	

Order 4, WSO 3, L-stable:
(DIRK436)

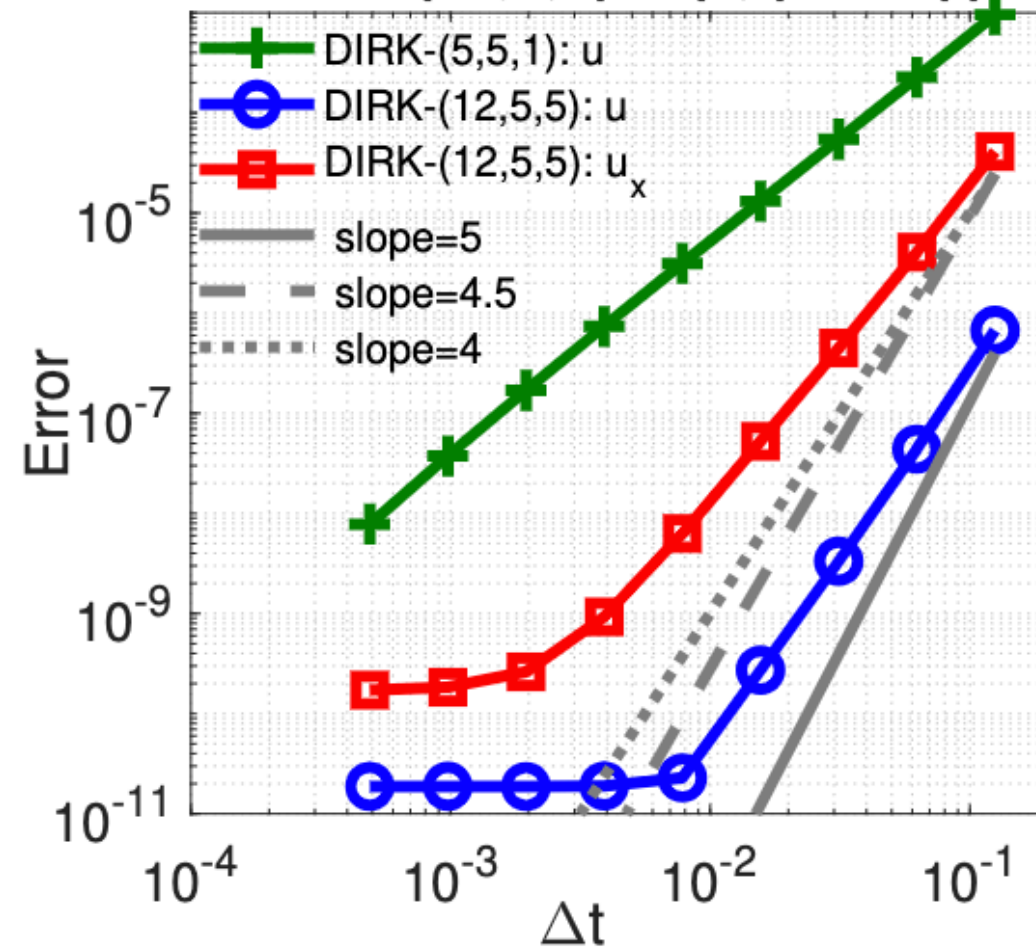
0.080		0.080					
0.464		0.328	0.136				
1.349		-0.651	1.743	0.256			
1.313		-0.715	1.794	-0.078	0.312		
0.989		-1.120	1.983	3.117	-3.762	0.771	
1.000		0.215	0.536	0.154	-0.218	0.072	0.240
			0.215	0.536	0.154	-0.218	0.072 0.240

Heat Eqn: DIRK-(7,4,4)



Viscous Burgers Eqn

DIRK-(12,5,5): $u(x,t)=\cos(t)$

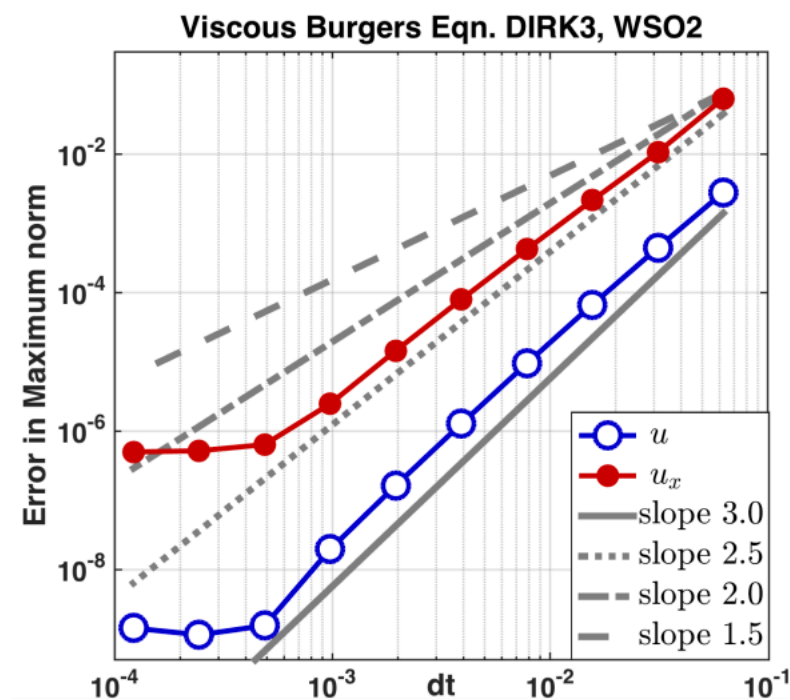
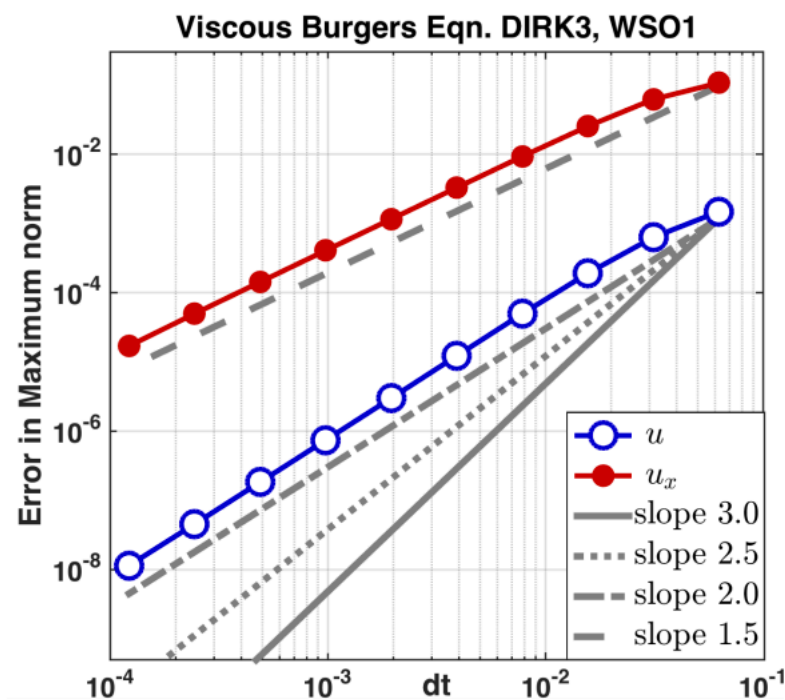


$$u_t + uu_x = \nu u_{xx} + f$$

Nonlinear PDE example: viscous Burgers' equation

$$u_t + uu_x = \frac{1}{10} u_{xx}$$

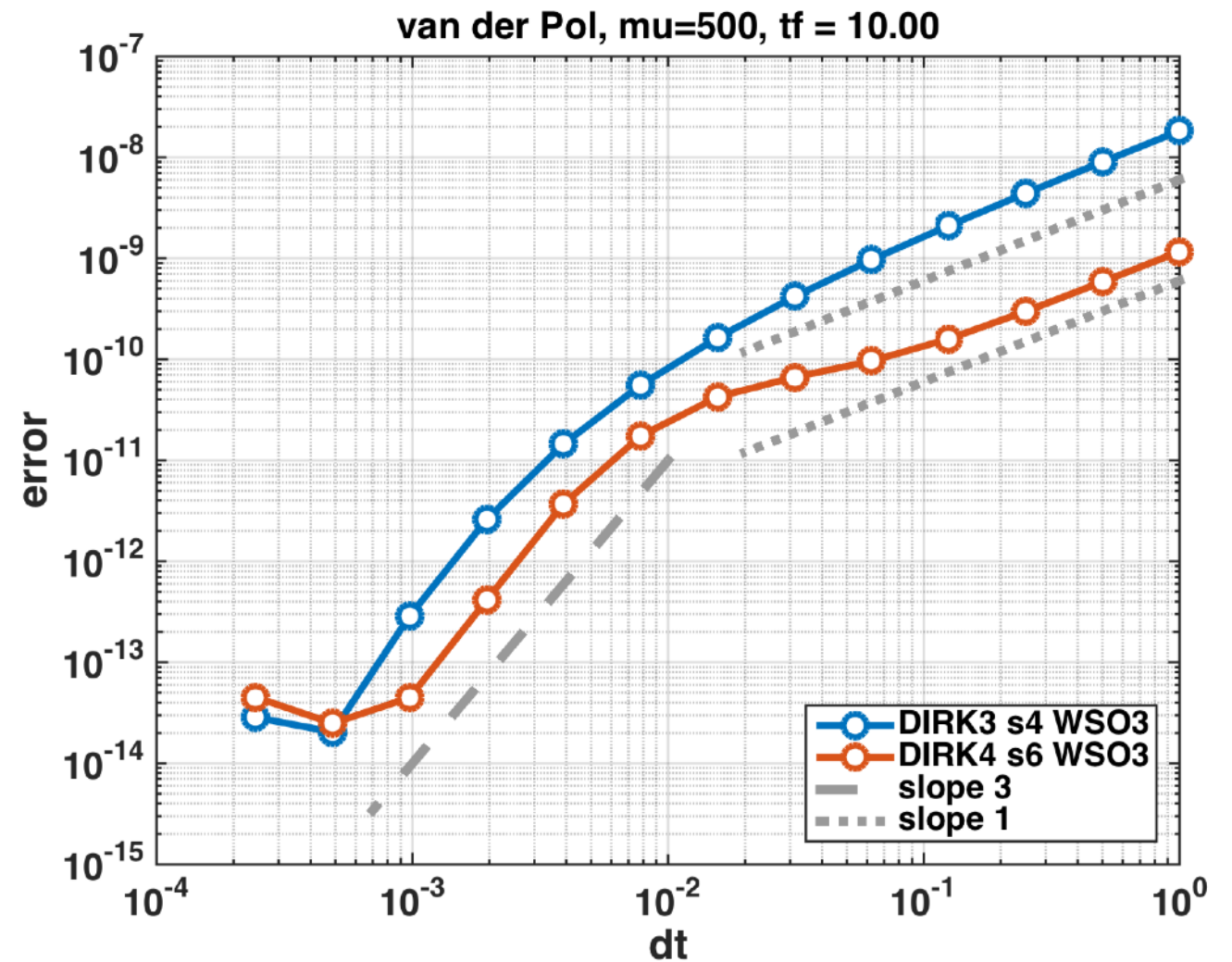
Manufactured solution with
time-dependent Dirichlet
boundary conditions



Observed convergence rate is again
 $\min(p, q+1)$ for schemes with weak
stage order q

Nonlinear ODE example: van der Pol oscillator

Weak stage order does not
mitigate order reduction for stiff
nonlinear ODEs



Order reduction from
boundary conditions

Simulating a flood

$$\begin{aligned} h_t + (hu)_x + (hv)_y &= 0 \\ (hu)_t + \left(hu^2 + \frac{1}{2}gh^2\right)_x + (huv)_y &= 0 \\ (hv)_t + (huv)_x + \left(hv^2 + \frac{1}{2}gh^2\right)_y &= 0 \end{aligned}$$

Space: 5th-order (Clawpack)

Time: 4th-order Runge-Kutta

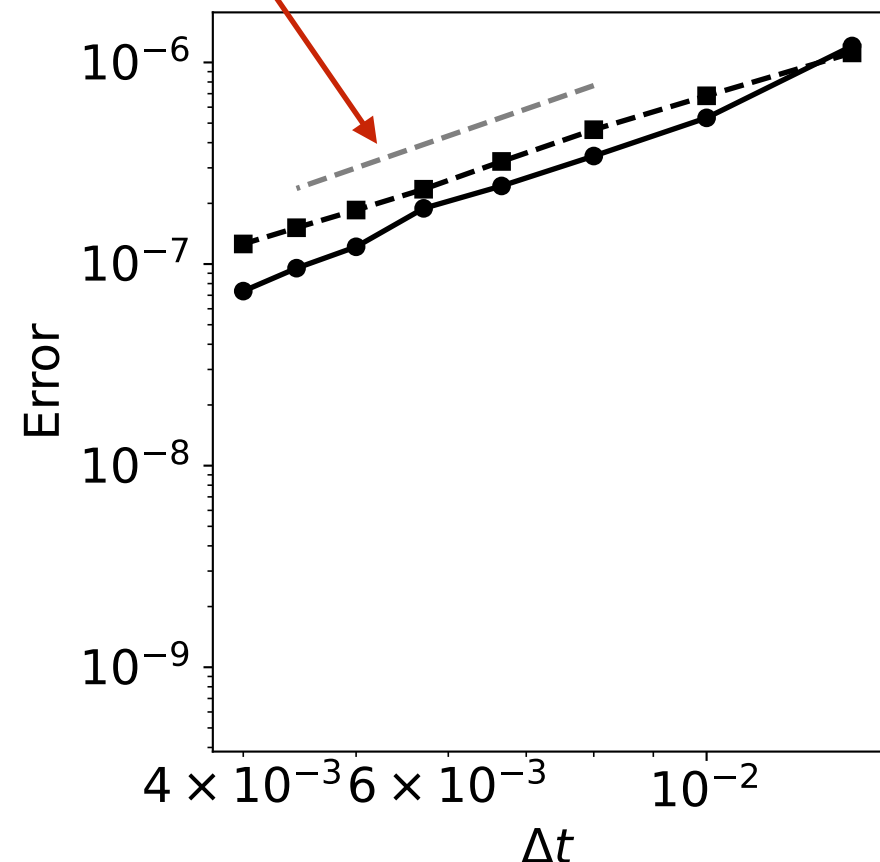
$$h(x, y, t) = \eta \left[\frac{T^2}{t^2 + T^2} - \frac{x^2 + y^2}{R_0^2} \left(\frac{T^2}{t^2 + T^2} \right)^2 \right]$$

$$u(x, y, t) = \frac{xt}{t^2 + T^2}$$

$$v(x, y, t) = \frac{yt}{t^2 + T^2}$$

Thacker (1981)

Only 2nd order convergence!



$$Y_1 = y_n$$

$$Y_2 = y_n + \frac{\Delta t}{2} f(Y_1)$$

$$Y_3 = y_n + \frac{\Delta t}{2} f(Y_2)$$

$$Y_4 = y_n + \Delta t f(Y_3)$$

$$y_{n+1} = y_n + \frac{\Delta t}{6} (f(Y_1) + 2f(Y_2) + 2f(Y_3) + f(Y_4))$$

$$Y_1 = y_n \approx y(t_n)$$

$$Y_2 = y_n + \frac{\Delta t}{2} f(Y_1) \approx y(t_n + \Delta t/2) \quad \text{1st order}$$

$$Y_3 = y_n + \frac{\Delta t}{2} f(Y_2) \approx y(t_n + \Delta t/2) \quad \text{1st order}$$

$$Y_4 = y_n + \Delta t f(Y_3) \approx y(t_n + \Delta t) \quad \text{2nd order}$$

$$y_{n+1} = y_n + \frac{\Delta t}{6} (f(Y_1) + 2f(Y_2) + 2f(Y_3) + f(Y_4)) \quad \text{4th order}$$

The method is designed so that the errors in the stages will exactly cancel up to a certain order

Order reduction is the failure of this cancellation

**THE CORRECT FORMULATION OF INTERMEDIATE BOUNDARY
CONDITIONS FOR RUNGE–KUTTA TIME INTEGRATION
OF INITIAL BOUNDARY VALUE PROBLEMS***

D. PATHRIA[†]

“A RK integrator with a high order of accuracy...
will very likely lose this unless the intermediate
boundary data exhibits the same time errors as
the intermediate stage values of the RK method
are designed to do.”

THE THEORETICAL ACCURACY OF RUNGE–KUTTA TIME DISCRETIZATIONS FOR THE INITIAL BOUNDARY VALUE PROBLEM: A STUDY OF THE BOUNDARY ERROR*

MARK H. CARPENTER[†], DAVID GOTTLIEB[‡], SAUL ABARBANEL[§], AND WAI-SUN DON[¶]

Abstract. The conventional method of imposing time-dependent boundary conditions for Runge–Kutta time advancement reduces the formal accuracy of the space–time method to first-order locally, and second-order globally, independently of the spatial operator. This counterintuitive result is analyzed in this paper.

Two methods of eliminating this problem are proposed for the linear constant coefficient case.

1. Impose the exact boundary condition only at the end of the complete Runge–Kutta cycle.
2. Impose consistent intermediate boundary conditions derived from the physical boundary condition and its derivatives.

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Convergence and Order Reduction of Runge-Kutta Schemes Applied to Evolutionary Problems in Partial Differential Equations

J.M. Sanz-Serna¹, J.G. Verwer², and W.H. Hundsdorfer²

- Rewrite problem as inhomogeneous PDE with homogeneous boundary conditions

Bounds for explicit WSO methods

$$p \leq \deg(R(z)) \leq \dim(Y)$$

$$\dim(Y) + \dim(K_q) \leq s$$

$$p \leq \deg(R(z)) \leq \dim(Y) \leq s - \dim(K_q)$$

$$q \leq \dim(K_q) + 1$$

$$p + q \leq s + 1$$

q and $R(z)$

Generally, explicit RK methods have

$$\deg(R(z)) = s$$

so

$$q = 1$$

To increase q , we must use more stages in a way that doesn't increase the degree of the stability polynomial!

The resulting methods are reducible when applied to linear, constant-coefficient problems.

$$R(z) = 1 + z + \frac{1}{2}z^2 + \dots + \frac{1}{p!}z^p + \dots + 0z^{s-q+1} + \dots + 0z^s$$

Last $q-1$ coefficients fixed by WSO q
 First $p+1$ coefficients fixed by classical order p
 Free coefficients

If $s = p+q-1$, the entire stability polynomial is determined.

Methods with $p+q = s+1$

$(3,2,2)$: 2-parameter family

0	
c_2	c_2
c_3	c_3
	$1 - \frac{1}{2c_2} - \frac{1}{2c_3}$
	$\frac{c_3}{2c_2(c_3 - c_2)}$
	$\frac{c_2}{2c_3(c_2 - c_3)}$

, $(3, 2, 2)$

0	
$\frac{1}{2}$	$\frac{1}{2}$
1	1
	$-\frac{1}{2}$
	2
	$-\frac{1}{2}$

Min. error

Methods with $p+q = s+1$

$(5,3,3)$: 6-parameter family

0					
$\frac{3}{11}$	$\frac{3}{11}$				
$\frac{15}{19}$	$\frac{285645}{493487}$	$\frac{103950}{493487}$			
$\frac{5}{6}$	$\frac{3075805}{5314896}$	$\frac{1353275}{5314896}$			
1	$\frac{196687}{177710}$	$-\frac{129383023}{426077496}$	$\frac{48013}{42120}$	$-\frac{2268}{2405}$	
	$\frac{5626}{4725}$	$-\frac{25289}{13608}$	$\frac{569297}{340200}$	$\frac{324}{175}$	$-\frac{13}{7}$

$(5,3,3)$

Min. error

0					
$\frac{1}{3}$	$\frac{1}{3}$				
$\frac{2}{3}$	$\frac{2}{3}$				
1	1				
0	$-\frac{11}{12}$	$\frac{3}{2}$	$-\frac{3}{4}$	$\frac{1}{6}$	
	$\frac{1}{4}$	-3	$\frac{15}{4}$	-1	1

ERK313

Skvortsov

In the paper, additional schemes are derived up to $(9,5,5)$

Minimum # of stages for order p and WSO q

$q \backslash p$	2	3	4	5
1	2	3	4	6
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8
5	6	7	8	9

Numerical Tests

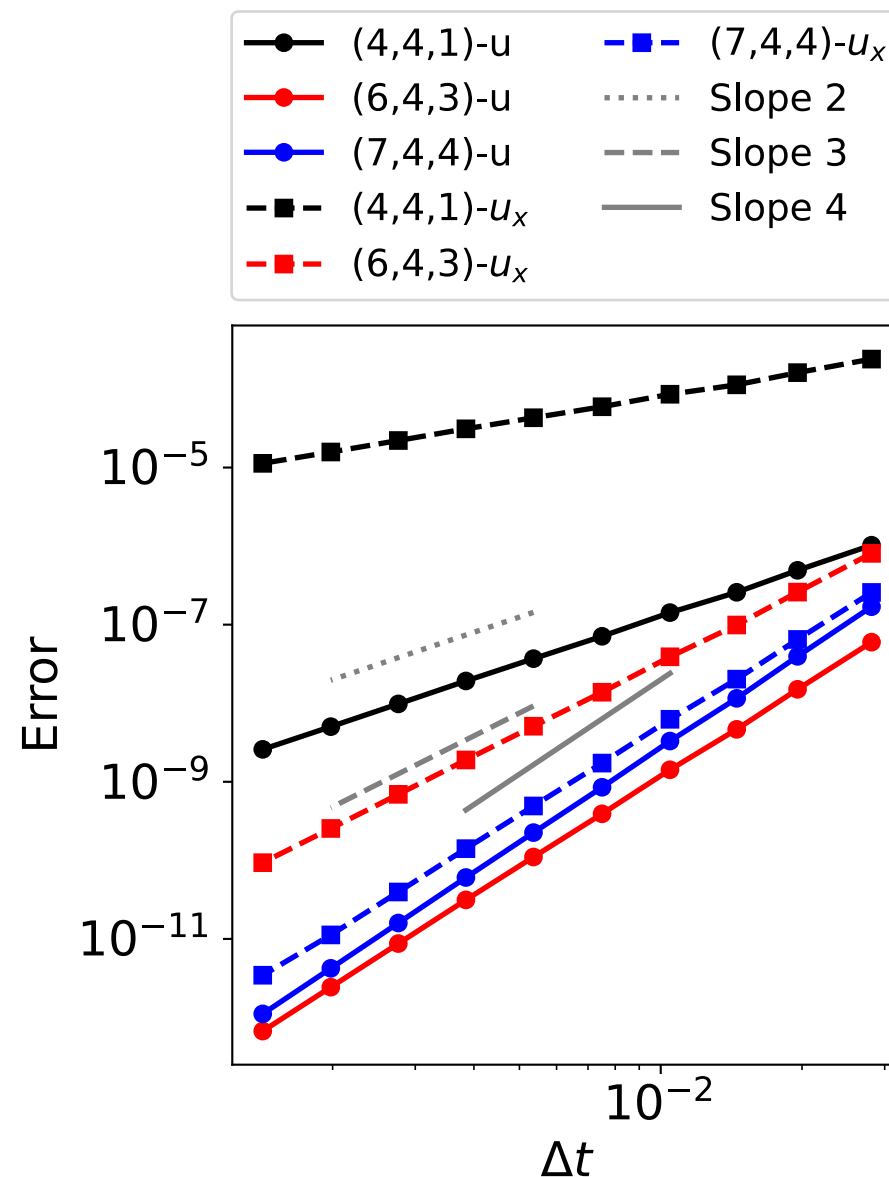
Advection

$$\begin{aligned} u_t + a(x)u_x &= f(x, t) \\ u(x, 0) &= u_0(x) \\ u(0, t) &= g_0(t), \end{aligned}$$

$$a(x) = \frac{1}{3}(\sin x + 2)$$

$$u(x, t) = \frac{1+x}{1+t}$$

Space: 1st-order upwind



2D acoustics

$$p_t + K_0 u_x + K_0 v_y = f^p$$

$$u_t + \frac{1}{\rho_0} p_x = f^u$$

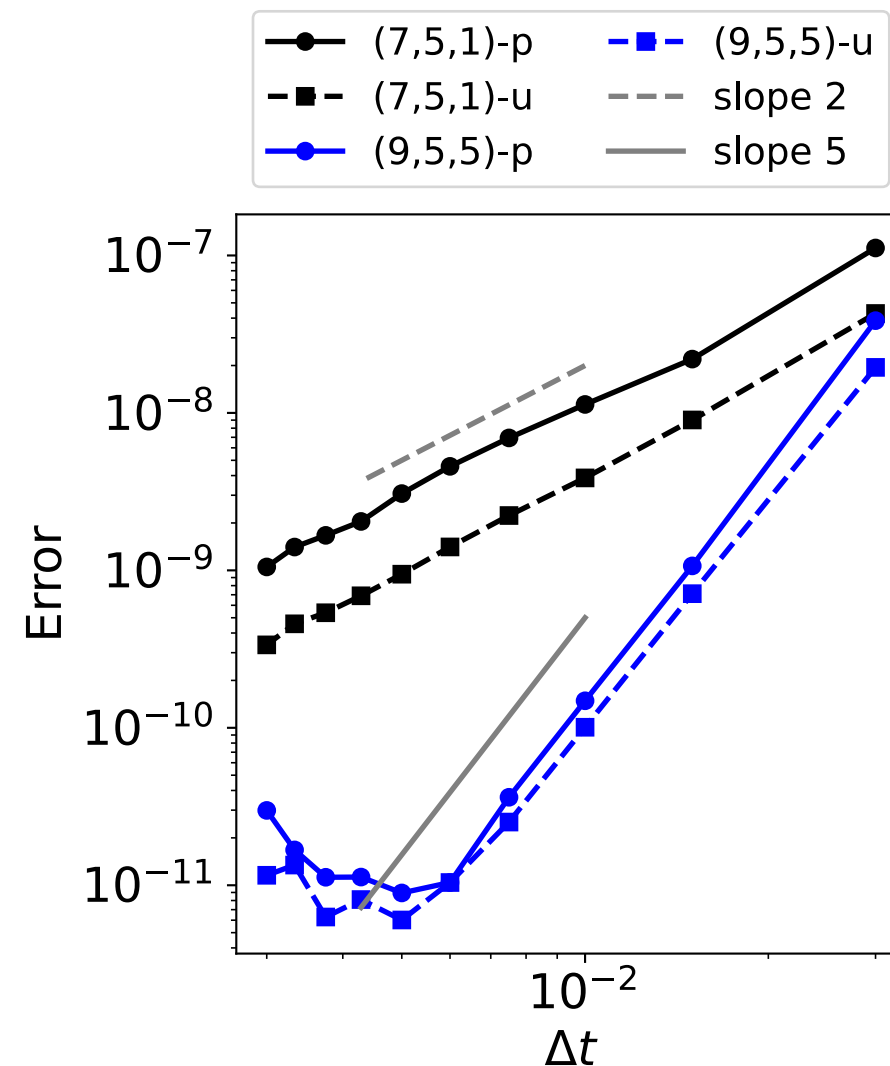
$$v_t + \frac{1}{\rho_0} p_y = f^v$$

$$p(x, y, t) = \frac{1 + x + y}{1 + t}$$

$$u(x, y, t) = \frac{xt}{1 + t}$$

$$v(x, y, t) = \frac{yt}{1 + t}$$

Space: 5th-order (Clawpack)



Burgers

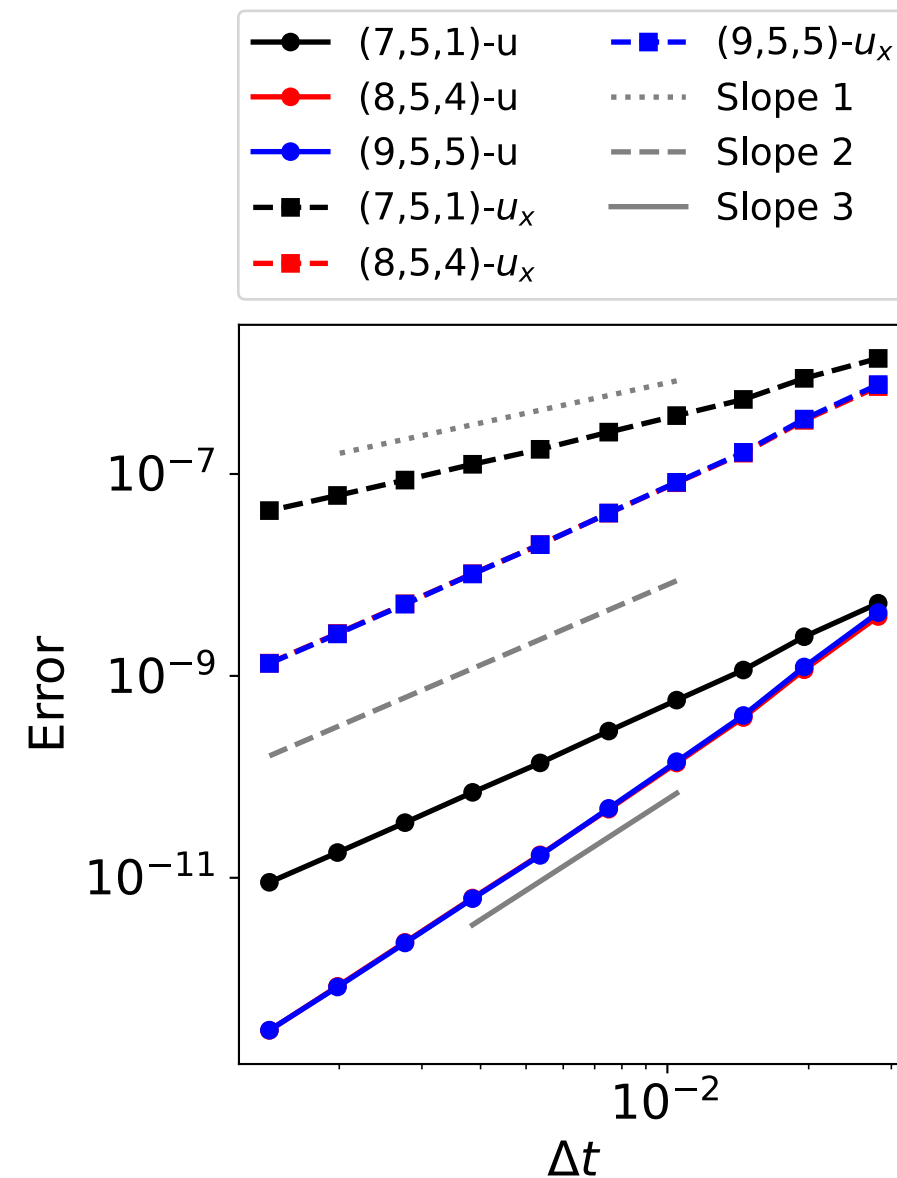
$$u_t + uu_x = 0$$

$$u(x, 0) = u_0(x)$$

$$u(0, t) = g_0(t)$$

$$u(x, t) = \frac{1+x}{1+t}$$

Space: 1st-order upwind



2D Shallow Water

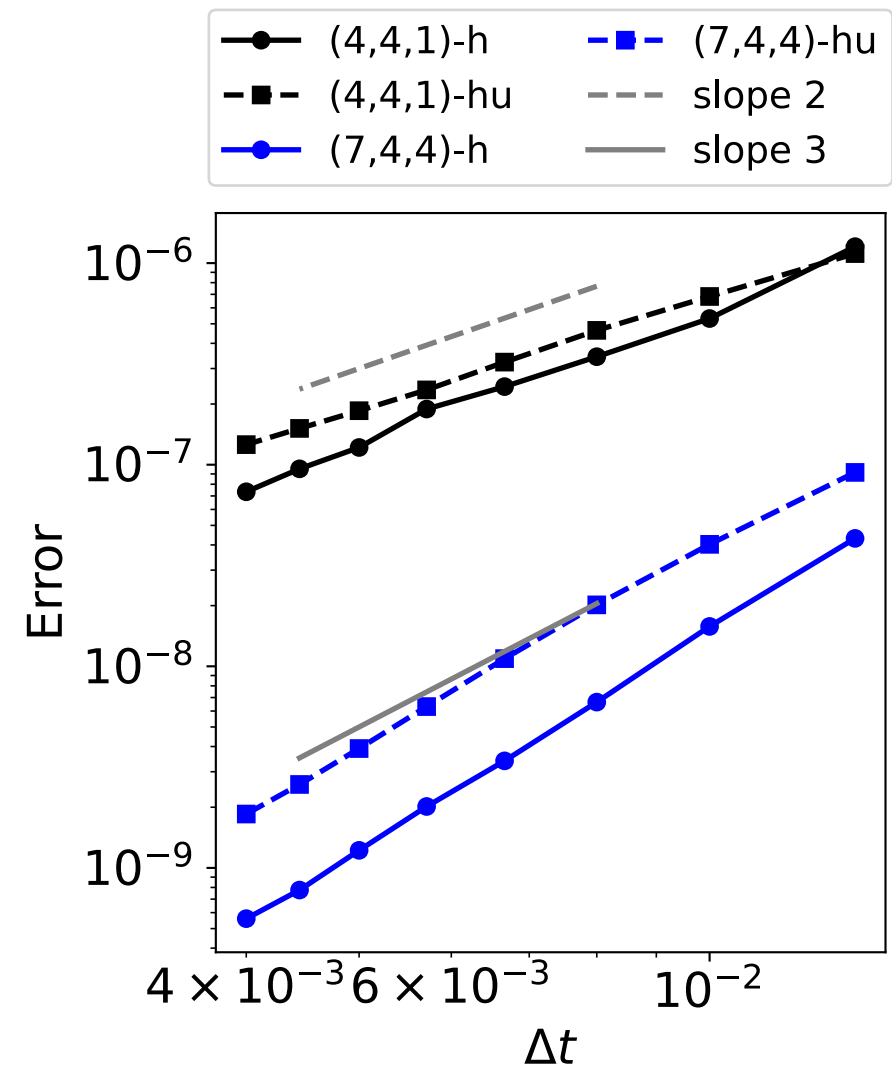
$$\begin{aligned} h_t + (hu)_x + (hv)_y &= 0 \\ (hu)_t + \left(hu^2 + \frac{1}{2}gh^2\right)_x + (huv)_y &= 0 \\ (hv)_t + (huv)_x + \left(hv^2 + \frac{1}{2}gh^2\right)_y &= 0 \end{aligned}$$

$$h(x, y, t) = \eta \left[\frac{T^2}{t^2 + T^2} - \frac{x^2 + y^2}{R_0^2} \left(\frac{T^2}{t^2 + T^2} \right)^2 \right]$$

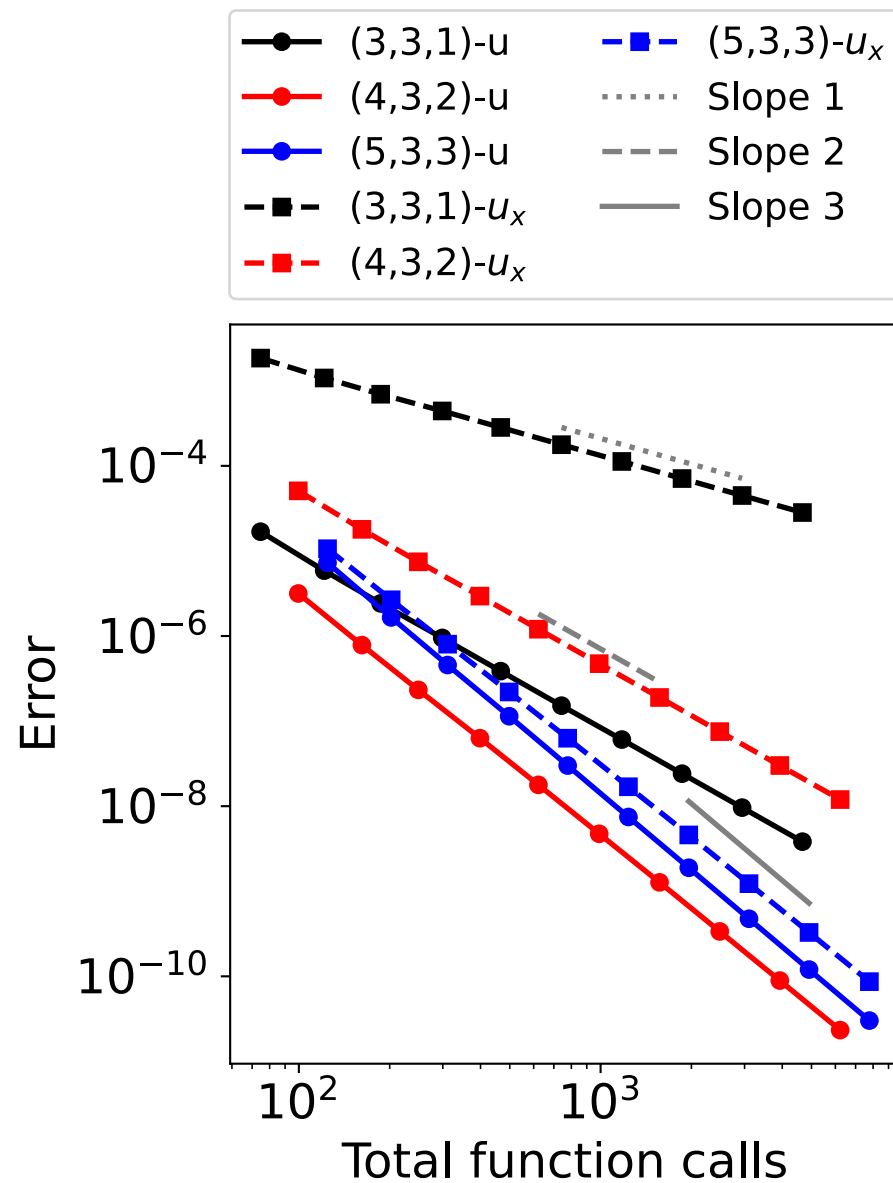
$$u(x, y, t) = \frac{xt}{t^2 + T^2}$$

$$v(x, y, t) = \frac{yt}{t^2 + T^2}$$

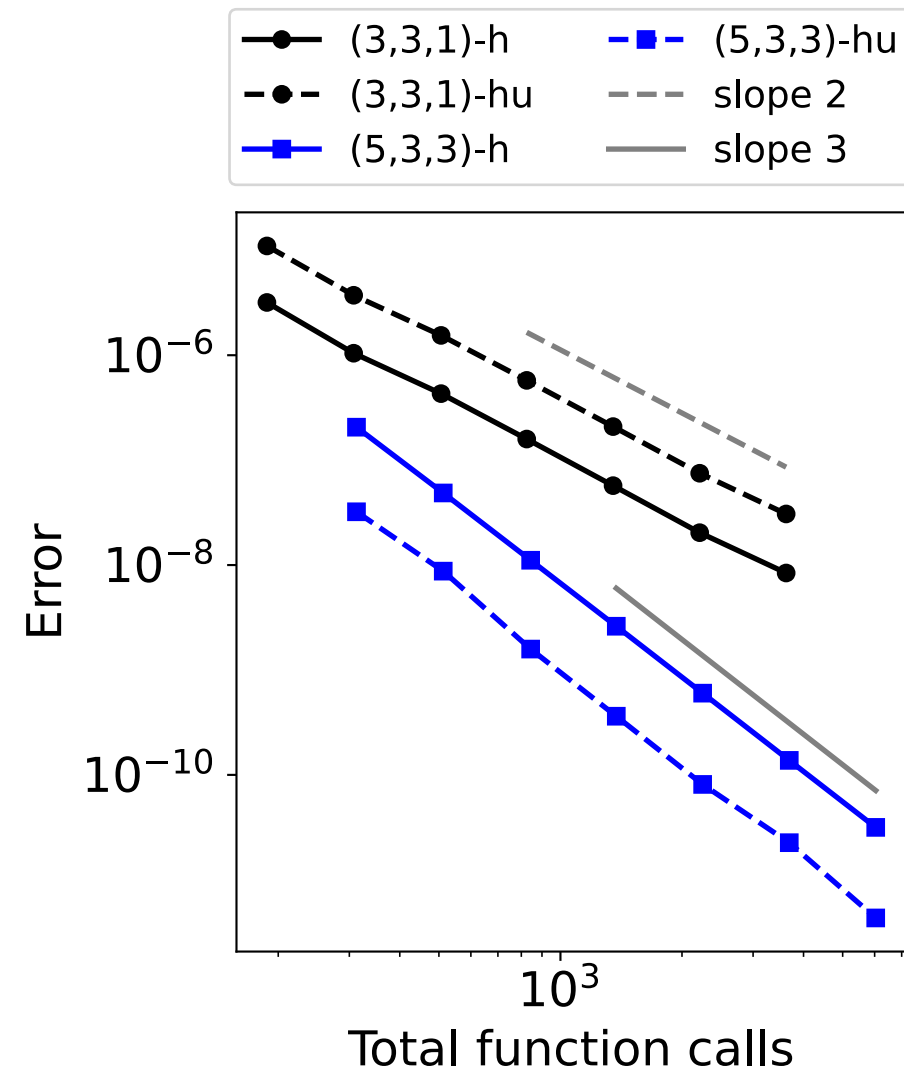
Space: 5th-order (Clawpack)



Work-precision comparison



1D Advection



1D Shallow water

Non-intrusive
implementation

Some comments on the
significance of order reduction

THE THEORETICAL ACCURACY OF RUNGE-KUTTA TIME DISCRETIZATIONS FOR THE INITIAL BOUNDARY VALUE PROBLEM: A STUDY OF THE BOUNDARY ERROR*

MARK H. CARPENTER[†], DAVID GOTTLIEB[‡], SAUL ABARBANEL[§], AND WAI-SUN DON[¶]

At this point we comment on why the above predicted phenomena has not been observed previously by the practitioners of higher-order methods. From (21) we see that the leading error coefficient in the expression is proportional to $\lambda^4 \delta x$. For example, if D represents a fourth-order central difference operator with fourth-order boundary closures and if $u = e^{i(x-t)}$, then the error at the point next to the boundary becomes

$$(28) \quad u_t + u_x = \lambda^4 \delta x \left[\frac{19}{165888} + i \frac{(2304 - 19\lambda)}{995328} \delta x + \dots \right].$$

Thus if $\lambda \ll 1$ for a given δx it is possible that $\lambda^4 \delta x \leq (\delta t)^4$. However, if one refines the grid, or conversely, if one runs the computation at the allowable stable CFL ($\lambda = O(1)$), then the first-order error becomes apparent. A detailed study of this effect is now presented.

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Convergence and Order Reduction of Runge-Kutta Schemes Applied to Evolutionary Problems in Partial Differential Equations

J.M. Sanz-Serna¹, J.G. Verwer², and W.H. Hundsdorfer²

It is fair to say that in practical problems the negative effects caused by order reduction are likely to be less important than those stemming from other sources, such as errors in space, instabilities at boundaries, curved boundaries, etc. However, the understanding of this phenomenon is essential in situations where one is interested in higher order methods.

A B-series perspective on order reduction

$$u(t+h) = \sum_{\tau \in T} h^{|\tau|} \phi(\tau) \alpha(\tau) F_f(\tau)(u(t))$$

Normalization here: $\phi(\tau) = \gamma(\tau) \sum_j b_j \Phi_j(\tau)$

Order conditions: $\phi(\tau) = 1$

Traditional approach: enforce order conditions for all terms up to some power of h.

$$u(t+h) = \sum_{\tau \in T} h^{|\tau|} \phi(\tau) \alpha(\tau) F_f(\tau)(u(t))$$

For stiff problems we have a large parameter λ .

We let $z = h\lambda$ and reorganize the terms:

$$u(t+h) = \sum_{j=0}^{\infty} h^j \sum_{k=0}^{\infty} z^k S_{jk}$$

Now for B-convergence, we impose the order conditions up to some power of h , but for ALL powers of z .

Example: Prothero-Robinson

$$y'(t) = \lambda(y - g(t)) + g'(t)$$

Rewrite as autonomous system of 2 equations and expand in a B-series:

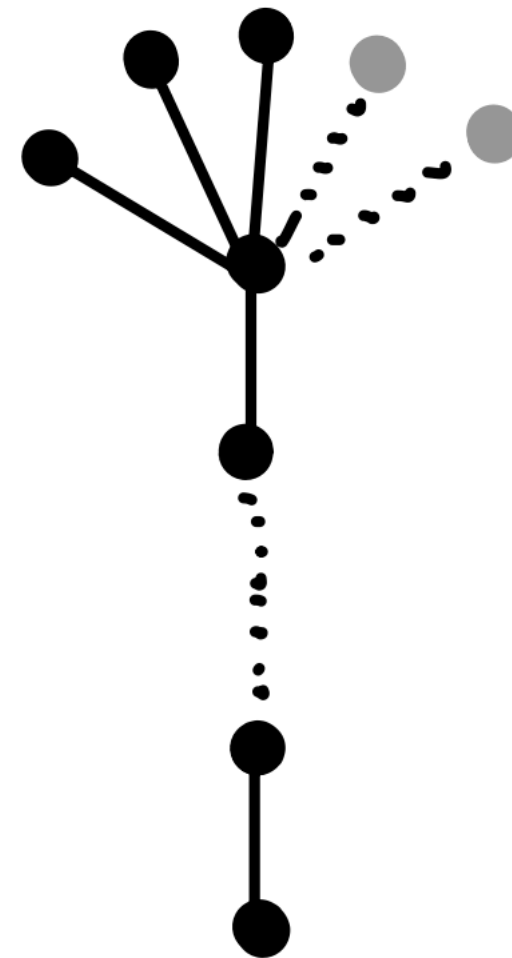
$$u_1'(t) = \lambda(u_1 - g(u_2)) + g'(u_2)$$

$$u_2'(t) = 1$$

$$\begin{aligned} y(t+h)|_{y(t)=g(t)} = & y + h\phi(\bullet)y' + h^2\phi(\bullet\bullet)y'' + h^3 \left[\lambda\phi(\bullet\bullet\bullet)y'' + \phi(\bullet\bullet\bullet)(y''' - \lambda y'') \right] \\ & + h^4 \left[\phi(\bullet\bullet\bullet\bullet)\lambda^2 y'' + \phi(\bullet\bullet\bullet\bullet)(\lambda y''' - \lambda^2 y'') + \phi(\bullet\bullet\bullet\bullet)(y^{(4)} - \lambda y''') \right] + O(h^5). \end{aligned}$$

$$y(t+h)|_{y(t)=g(t)} = y + h\phi(\bullet)y' + h^2\phi(\text{⦿})y'' + h^3 \left[\lambda\phi(\text{⦿})y'' + \phi(\text{⦿})'(y''' - \lambda y'') \right] \\ + h^4 \left[\phi(\text{⦿})\lambda^2 y'' + \phi(\text{⦿})'(\lambda y''' - \lambda^2 y'') + \phi(\text{⦿})''(y^{(4)} - \lambda y''') \right] + O(h^5).$$

- In this case, all differentials can be expressed simply as derivatives of y
- All trees that appear are “palm trees”:



$$\begin{aligned}
y(t+h)|_{y(t)=g(t)} = & y + \boxed{h\phi(\bullet)y'} + \boxed{h^2\phi(\bullet\bullet)y''} + h^3 \left[\boxed{\lambda\phi(\bullet\bullet)y''} + \boxed{\phi(\bullet\bullet\bullet)(y''')} - \boxed{\lambda y''} \right] \\
& + h^4 \left[\boxed{\phi(\bullet\bullet\bullet)\lambda^2 y''} + \boxed{\phi(\bullet\bullet\bullet\bullet)(\lambda y''')} - \boxed{\lambda^2 y''} \right] + \boxed{\phi(\bullet\bullet\bullet\bullet)(y^{(4)})} - \boxed{\lambda y'''} \Big] + O(h^5).
\end{aligned}$$

- We have all the traditional “bushy-tree” (quadrature) conditions, with no λ 
- The remaining conditions appear in pairs   

$$\begin{aligned}
y(t+h)|_{y(t)=g(t)} = & y + h\phi(\bullet)y' + h^2\phi(\text{⦿})y'' + h^3 \left[\lambda\phi(\text{⦿})y'' + \phi(\text{⦿}) (y''' - \lambda y'') \right] \\
& + h^4 \left[\phi(\text{⦿})\lambda^2 y'' + \phi(\text{⦿}) (\lambda y''' - \lambda^2 y'') + \phi(\text{⦿}) (y^{(4)} - \lambda y''') \right] + O(h^5).
\end{aligned}$$

Rewriting in terms of z , we find:

$$y(t+h)|_{y(t)=g(t)} = y + h\phi(\bullet)y' + \sum_{j=2}^{\infty} h^j y^{(j)} \left[\phi(\tau_{1,j}) + \sum_{k=1}^{\infty} z^k (\phi(\tau_{k+1,j}) - \phi(\tau_{k,j+1})) \right]$$

So we require

$$\begin{aligned}
\phi(\tau_{1,j}) &= 1 & 1 \leq j \leq q \\
\phi(\tau_{k+1,j}) - \phi(\tau_{k,j+1}) &= 0 & 1 \leq j \leq q, \quad 1 \leq k \leq \infty.
\end{aligned}$$

These are just the “weak stage order” conditions.